

A37 TUT17

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March 29, 2026

Sequences, BMCT, and Series

Coverage

Definition 1 (Bounded). A sequence $\{a_n\}$ is bounded if it is both bounded above and bounded below, i.e.

$$\exists C \in \mathbb{R}^+ \cup \{0\} \text{ s.t. } |a_n| \leq C \quad \forall n \in \mathbb{N} \Leftrightarrow -C \leq a_n \leq C$$

Theorem 1 (BMCT). Suppose $\{a_n\}$ bounded and monotone, in particular:

- $\{a_n\}$ is (strictly) increasing AND bounded above

OR:

- $\{a_n\}$ is (strictly) decreasing AND bounded below.

Then $\{a_n\}$ converges.

Definition 2 (Series). Let $\{a_n\}$ be a sequence, then the (infinite) series is the formal sum:

$$a_1 + a_2 + \dots + a_n + \dots = \sum_{n=1}^{\infty} a_n$$

where:

- a_n is the general term of $\sum a_n$
- $\forall n \in \mathbb{N}$, the n -th partial sum of $\sum a_n$ is defined as

$$S_n = a_1 + a_2 + \dots + a_n$$

Remark. $\sum a_n \approx S_n$

Definition 3 (Series Convergence). A series a_n converges if

$$\lim_{n \rightarrow \infty} S_n = s, \quad s \in \mathbb{R}$$

We may write:

$$\sum_{n=1}^{\infty} a_n = \lim_{n \rightarrow \infty} S_n$$

If it does not converge, then the series diverges.

Proposition 1. Let $\sum a_n$, $\sum b_n$ be convergent series, then the following properties hold:

- linearity
- Vanishing condition: $\lim_{n \rightarrow \infty} a_n = 0$

Exercises

Let us first do a classic example:

Example 1. Prove that $\{\sqrt[n]{n}\}$ converges to 1.

Solution. There are multiple ways to approach this question. We will focus on ε -N proof, and utilize AM-GM inequality. Clearly, $l = 1$.

Recall:

$$\sqrt[n]{a_1 \dots a_n} \leq \frac{a_1 + \dots + a_n}{n}$$

where the first is the geometric mean and the second is the corresponding arithmetic mean.

Let $\varepsilon > 0$ be arbitrary. Choose $N = \frac{4}{\varepsilon^2}$. Suppose $n \in \mathbb{N}^\times$, $n > N$.

$$\begin{aligned} |a_n - 1| &= |\sqrt[n]{n} - 1| = \sqrt[n]{n} - 1 \\ &= \sqrt[n]{1 \cdot 1 \cdot \dots \cdot 1 \cdot \sqrt{n} \cdot \sqrt{n} - 1} \\ &\leq \frac{1 + 1 + \dots + 1 + 2\sqrt{n} - n}{n} = \frac{n - 2 + 2\sqrt{n} - n}{n} = \frac{2\sqrt{n} - 2}{n} \\ &\leq \frac{2\sqrt{n}}{n} = \frac{2}{\sqrt{n}} \\ &\leq \frac{2}{\sqrt{N}} \\ &= \frac{2}{\sqrt{\frac{4}{\varepsilon^2}}} \\ &= \varepsilon \end{aligned}$$

□

(You can definitely fill in the justifications yourself at this point :D)

□

The following examples walk us through how to utilize helper assumption to bound “ $|a_n - b|$ ”-ish expressions in an $\varepsilon - N$ proof.

Example 2. Prove that $\frac{n^2}{n^2+n-6}$ converges with and without helper assumptions.

Solution. Clearly by performing simple analysis, having a leading term of $\frac{n^2}{n^2} = 1$ implies that as $n \rightarrow \infty$ the sequence is eventually going to 1. We deduce that $l = 1$.

Case (i): exercise - or email me for help.

Case (ii): Let $\varepsilon > 0$ be arbitrary. Choose $N = \max\{6, \frac{1}{\varepsilon}\} > 0$. Suppose $n \in \mathbb{N}^\times$, $n > N$.

$$\begin{aligned}
|a_n - 1| &= \left| \frac{n^2}{n^2 + n - 6} - 1 \right| && \text{(By def.)} \\
&= \left| \frac{-n + 6}{n^2 + n - 6} \right| && \text{(Simplification)} \\
&= \left| \frac{n - 6}{n^2 + n - 6} \right| && \text{(By abs. value prop.)} \\
&= \frac{|n - 6|}{|n^2 + n - 6|} && \text{(By abs. value prop.)} \quad (*)
\end{aligned}$$

This is where we run into the problem: we cannot determine whether it is positive or negative for both the numerator and denominator. To resolve this, we must make a “helper assumption” which forces us to stay away from a place where we couldn’t determine the sign of our expression.

Note $n - 6 > 0 \Leftrightarrow n > 6$, then we let $n > 6$. With this assumption on n , we can arrive at the conclusion which $|n - 6| = n - 6$. We can also conclude that: $n + 3 > 0$, and $n - 2 > 0$.

$$\begin{aligned}
\therefore (*) &= \frac{n - 6}{n^2 + n - 6} \\
&\leq \frac{n}{n^2 + n - 6} && \text{(i.e. we max num.)} \\
&\leq \frac{n}{n^2} && (\because n - 6 > 0 \Rightarrow \frac{1}{n^2} \geq \frac{1}{n^2 + n - 6}) \\
&= \frac{1}{n} < \frac{1}{N} = \frac{1}{\frac{1}{\varepsilon}} = \varepsilon
\end{aligned}$$

Note: by taking the maximum of the two, we effectively avoid the potential problem that ε was taken to be arbitrary - the max covers both cases perfectly. $\square$

Example 3. Prove that $a_n = \frac{\arctan(n)}{7n-9}$ converges.

Solution. By simple reasoning, we deduce that as $n \rightarrow \infty$, $a_n \rightarrow 0 \Rightarrow l = 0 \in \mathbb{R}$.

Let $\varepsilon > 0$ be arbitrary. Choose $N = \max\{2, \frac{5\varepsilon}{\pi}\} > 0$. Suppose $n \in \mathbb{N}^\times$, $n > N$.

$$\begin{aligned}
\left| \frac{\arctan(n)}{7n-9} - 0 \right| &= \left| \frac{\arctan(n)}{7n-9} \right| \\
&= \frac{|\arctan(n)|}{|7n-9|} && \text{(By abs. value prop.)} \\
&= \frac{\frac{\pi}{2}}{|7n-9|} && \text{(By arctan prop.)} \quad (*)
\end{aligned}$$

Again, we run into a problem here - we cannot bound the denominator in any effective way. Therefore, let $n > 2 \Rightarrow \frac{1}{n} < \frac{1}{2} \Leftrightarrow \frac{9}{n} < \frac{9}{2} \Leftrightarrow 7 - \frac{9}{n} > 7 - \frac{9}{2} = \frac{5}{2} \Leftrightarrow \frac{1}{7 - \frac{9}{n}} < \frac{1}{\frac{5}{2}} = \frac{2}{5}$.

Hence, to rewrite (*), we get...

$$\begin{aligned}
(*) &= \frac{\pi}{2} \cdot \frac{1}{|n|} \cdot \frac{1}{|7 - \frac{9}{n}|} \\
&= \frac{\pi}{2} \cdot \frac{1}{n} \cdot \frac{1}{7 - \frac{9}{n}} \quad (\because n > 2 \Rightarrow |n| > 0; |\frac{1}{7 - \frac{9}{n}}| = \frac{1}{7 - \frac{9}{n}}) \\
&\leq \frac{\pi}{2} \cdot \frac{1}{N} \cdot \frac{2}{5} \\
&= \frac{\pi}{5N} \\
&= \frac{\pi}{5} \cdot \frac{5\varepsilon}{\pi} = \varepsilon
\end{aligned}$$

□

Example 4. Prove that $a_n = \frac{\sin(\sqrt{n})}{2n-5}$

Solution. Again, clearly it converges to 0.

Let $\varepsilon > 0$ be arbitrary. Choose $N = \max\{3, \frac{3}{\varepsilon}\} > 0$. Suppose $n \in \mathbb{N}^\times$, $n > N$.

$$\begin{aligned}
\Rightarrow |a_n - l| &= \left| \frac{\sin(\sqrt{n})}{2n-5} \right| \\
&\leq \frac{1}{|2n-5|} \\
&= \frac{1}{n} \cdot \frac{1}{|2 - \frac{5}{n}|} \\
&= \frac{1}{n} \cdot \frac{1}{2 - \frac{5}{n}} \quad (\text{why? what } n \text{ should I choose?}) \\
&< \frac{1}{N} \cdot 3 \\
&= \frac{1}{\frac{3}{\varepsilon}} \cdot 3 \\
&= \varepsilon
\end{aligned}$$

(Fill in the justifications.)

□

Example 5. Prove that $a_n = \frac{4n^2-7}{2n^3-5}$ converges.

Solution. Clearly, $l = 0$.

Let $\varepsilon > 0$ be arbitrary. Choose $N = \max\{3, \frac{108}{49}\varepsilon\} > 0$. Suppose $n \in \mathbb{N}^\times$, $n > N$.

$$\begin{aligned}
\Rightarrow |a_n - l| &= \frac{4n^2 - 7}{2n^3 - 5} \\
&= \frac{|4n^2 - 7|}{|2n^3 - 5|} \\
&\leq \frac{4n^2 - 7}{2n^3 - 5} \quad (\text{By given: } n > 2) \\
&\leq \frac{4n^2}{2n^3 - 5} \\
&= \frac{4}{2n - \frac{5}{n^3}} \\
&= 4 \cdot \frac{1}{n} \cdot \frac{1}{2 - \frac{5}{n^3}} \quad (\text{let } n > 3) \\
&= \frac{4}{n} \cdot \frac{27}{49} \quad (\text{how?}) \\
&< \frac{108}{49N} \\
&= \frac{108}{49} \cdot \frac{1}{\frac{108}{49\epsilon}} \\
&= \epsilon
\end{aligned}$$

□

Example 6. Determine whether the Fibonacci sequence is divergent, where

$$F_0 = 0, F_1 = 1, F_n = F_{n-1} + F_{n-2} \quad \forall n > 1$$

Solution. We first make an educated guess on whether the sequence converges or diverges. It is trivial to see that by the construction of Fibonacci sequence, it will eventually diverge as $n \rightarrow \infty$. If we use Binet's formula, we can perform an N-M proof to show divergence. However, let's not rely on Binet's, so what tools do we have to show divergence of the Fibonacci sequence?

We will employ the contrapositive of "convergent sequences are bounded" (i.e. if a_n is unbounded then it diverges.) That is to say, we want to find a lower bound for sufficiently large n that this lower bound diverges eventually, this can be generally done by induction.

WTS: by induction, $F_n \geq n - 2, \forall n \geq 1$.

Base Case: we verify that the inequality holds for small values of n .

$\Rightarrow$ the inequality trivially holds for $n = 1, 2, 3, 4$.

Inductive Step: WTS: $F_{k+2} \geq (k+2) - 2 = k$ (i.e. it holds for $n = k+2$)

- Suppose that $k \geq 3$, the inequality holds for all values up to $k+1$

$$F_n \geq n - 2, \forall n = 1, 2, \dots, k+1$$

Clearly: from the Fibonacci recurrence relation:

$$F_{k+2} = F_{k+1} + F_k$$

Using I.H.: $F_{k+1} \geq (k+1) - 2 = k - 1$, $F_k \geq k - 2$

Then $F_{k+2} = F_{k+1} + F_k \geq (k-1) + (k-2) = 2k - 3$

Since $k \geq 3$, we observe that $2k - 3 \geq k$

$\therefore F_{k+2} \geq k$, which shows that it is unbounded. Hence, the Fibonacci sequence diverges. $\square$

Example 7. Consider the following sequence, defined recursively:

$$a_1 = 1$$

$$a_{n+1} = 3 - \frac{1}{a_n}$$

Prove that the function is bounded above and increasing, then conclude that it converges using BMCT. Challenge: find the limit of a_n .

Solution. We use induction to prove both hypotheses.

Bounded above: it is clear to us that a valid choice would be 3 since for each term, we are subtracting a positive number from 3.

Base case: clearly, when $n = 1$, $a_1 = 1 < 3$.

Inductive Step: WTS: $a_{k+1} < 3$ for some $k \in \mathbb{N}$ i.e. $3 - \frac{1}{a_k} < 3$

I.H.: suppose $a_k < 3$ for some $k \in \mathbb{N}$

$$\begin{aligned} a_k &< 3 && \text{(By IH)} \\ \Leftrightarrow \frac{1}{a_k} &> \frac{1}{3} && \text{(Taking reciprocal of both sides)} \\ \Leftrightarrow -\frac{1}{a_k} &< -\frac{1}{3} \\ \Leftrightarrow 3 - \frac{1}{a_k} &< 3 - \frac{1}{3} < 3 \end{aligned}$$

Therefore: $3 - \frac{1}{a_k} = a_{k+1} < 3$ (as wanted).

By induction: a_n is bounded above by 3.

Increasing:

Base case: clearly, $a_1 = 1 < 2 = 3 - \frac{1}{1} = a_2$

Inductive step: WTS: $a_{k+1} < a_{k+2}$

I.H.: suppose $a_k < a_{k+1}$ for some $k \in \mathbb{N}$

$$a_k < a_{k+1} \quad (\text{By IH}) \quad (1)$$

$$\Leftrightarrow \frac{1}{a_k} > \frac{1}{a_{k+1}} \quad (2)$$

$$\Leftrightarrow -\frac{1}{a_k} < -\frac{1}{a_{k+1}} \quad (3)$$

$$\Leftrightarrow 3 - \frac{1}{a_k} < 3 - \frac{1}{a_{k+1}} \quad (4)$$

$$\Leftrightarrow a_{k+1} < a_{k+2} \quad (\text{By def. of } a_n) \quad (5)$$

By induction: a_n is increasing.

Therefore, it is bounded above and increasing $\Rightarrow$ by BMCT: a_n converges.

In order to determine its limit, we see that

$$\lim_{n \rightarrow \infty} a_n = l = \lim_{n \rightarrow \infty} \left(3 - \frac{1}{a_n}\right) = 3 - \lim_{n \rightarrow \infty} \frac{1}{a_n} = 3 - \frac{1}{l}$$

We obtain that:

$$l = 3 - \frac{1}{l} \Leftrightarrow l^2 - 3l + 1 = 0$$

By using quadratic formula:

$$L_1 = \frac{3 + \sqrt{5}}{2}, \quad L_2 = \frac{3 - \sqrt{5}}{2}$$

To determine the correct limit, we make an observation that, by construction: a_n is bounded below by 1. (Why?)

Clearly $2 = \sqrt{4} < \sqrt{5}$ then $3 - \sqrt{5} < 3 - \sqrt{4} = 1$. Hence, $L_2 < \frac{1}{2} < 1$, which is impossible to be our limit. (Why?)

Therefore, as $n \rightarrow \infty$, $a_n \rightarrow L_1$. □

Series Convergence/Divergence

Coverage

In this week, we will discover whether a series converges or diverges. Thankfully, we will be equipped with the following tools to assist us in order to determine whether a series blows up to infinity or not.

Recall. The vanishing condition states that:

$$\sum a_n \text{ converges} \Rightarrow \lim_{n \rightarrow \infty} a_n = 0$$

Theorem 2 (Divergence Test). The divergence test is a criterion for a series diverging given by the vanishing condition:

$$\lim_{n \rightarrow \infty} a_n \neq 0 \Rightarrow \sum a_n \text{ diverges}$$

Definition 4 (Geometric Series). A G.S. is a particular series of the form

$$a + ar + ar^2 + \dots + ar^n + \dots = \sum_{n=0}^{\infty} ar^n$$

Where r is the common ratio of the series.

Solely from its definition, if we want to show a series is a G.S., we rewrite it in its specific form and identify what a and its common ratio r is. As one would expect, the G.S. is armed with a theorem to determine the convergence/divergence.

Theorem 3 (Geometric Series Test). Let $\sum ar^n$ be a G.S., then...

- i. if $|r| < 1$, $\sum ar^n$ converges and $\sum ar^n = \frac{a}{1-r}$.
- ii. if $|r| \geq 1$, $\sum ar^n$ diverges.

Your textbook has an amazing summary of all the tests you are allowed to use. I will briefly mention about them here again.

Theorem 4 (Integral Test). Given a series $\sum a_n$, define $f(n) = a_n$. If $f(x)$ is continuous, positive and decreasing on $[k, \infty)$, then...

$$\sum_{n=k}^{\infty} a_n \text{ converges} \Leftrightarrow \int_k^{\infty} f(x) dx \text{ converges}$$

Note that when we use IT, we define the function f on the real numbers but based on our sequence a_n . To determine whether f is increasing or decreasing, take its derivative. When proving the corresponding integral converges, calculate the integral **by definition**.

Remark. Including but not limited to IT, most of the convergence test will not tell you the sum of the series.

Definition 5 (*p-series*). A *p-series* is a series of the form $\sum_{n=1}^{\infty} \frac{1}{n^p}$ where $p \in \mathbb{R}^+$.

Theorem 5 (*p-series Test*). Given a *p-series* $\sum_{n=1}^{\infty} \frac{1}{n^p}$, then...

- i. If $p > 1$, then the series converges.
- ii. If $p \in (0, 1]$, then the series diverges.

Remark. To use *p-series* test, you must show that you have a *p-series* indeed i.e. identify the value of p .

Note that the following theorem is CT for improper integrals is a generalized version of the following:

Theorem 6 (*Comparison Theorem (CT)*). Let $\sum a_n$, $\sum b_n$, $\sum c_n$ be series. Just like CT for improper integrals, we also have two cases:

- i. If $(0 \leq a_n \leq b_n \ \forall n \in \mathbb{N}) \wedge (\sum b_n \text{ converges})$, then $\sum a_n$ converges.
- ii. If $(0 \leq c_n \leq a_n \ \forall n \in \mathbb{N}) \wedge (\sum c_n \text{ converges})$, then $\sum a_n$ diverges.

Definition 6 (*Alternating Series*). An alternating series is a series of the form $\sum (-1)^{n+1} b_n$ or $\sum (-1)^n b_n$, where $b_n > 0 \ \forall n \in \mathbb{N}$

Theorem 7 (*Alternating Series Test*). If b_n is decreasing and positive $\forall n$, and, $\lim_{n \rightarrow \infty} b_n = 0$, then

$$\sum (-1)^n b_n \text{ converges}$$

Remark. Unlike with IT, you can't take derivatives of b_n here to determine whether it is decreasing. The correct approach here is to use "bounds".

Definition 7 (*AC, CC*). A series $\sum a_n$ is...

- i. Absolutely convergent: if $\sum a_n$ converges and $\sum |a_n|$ converges.
- ii. Conditionally convergent: if $\sum a_n$ converges and $\sum |a_n|$ diverges.

Theorem 8 (*Ratio Test*). Let $\sum a_n$ be a series, $a_n \neq 0 \ \forall n \in \mathbb{N}$, define $\zeta = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right|$, then...

- if $\zeta < 1$, then $\sum a_n$ is absolutely convergent (and hence convergent).
- if $\zeta > 1$, then $\sum a_n$ is divergent.
- if $\zeta = 1$, the test is inconclusive, proceed with other tests.

Remark. Ratio Test is most commonly used on series with factorials or exponentials.

When to Use Which?

1. Does $\lim_{n \rightarrow \infty} a_n \neq 0$? If so, the series diverges by the Divergence Test. (Only apply this if its clear from inspection that the terms dont go to zero.)
2. Is it a p -series or geometric series?

$$\sum \frac{1}{n^p}, \quad \sum_{n=0}^{\infty} ar^n, \quad \sum_{n=1}^{\infty} ar^{n-1}$$

Use:

p -series converges if $p > 1$, geometric converges if $|r| < 1$

(You may need algebra to rewrite into standard form.)

3. Is it comparable to a p -series or geometric series? Try the Comparison Test.
4. Is it a rational expression (polynomials or radicals of polynomials)? Use the Comparison or Limit Comparison Test. (All terms must be positive.)
5. Does it involve factorials or powers like c^n ? Try the Ratio Test. (Factorials $\Rightarrow$ Ratio Test is usually required.)
6. Can it be written as

$$a_n = (-1)^n b_n \quad \text{or} \quad a_n = (-1)^{n+1} b_n?$$

Use the Alternating Series Test.

7. Can it be written as

$$a_n = (b_n)^n?$$

Try the Root Test.

8. Can you write $a_n = f(n)$ where f is positive and decreasing, and

$$\int_a^{\infty} f(x) dx$$

is manageable? Use the Integral Test.

Exercises

We first go through some examples of telescoping sum, it occurs the most with logarithm, so let us focus more on telescoping sum with PFD.

Example 8. For each of these series, determine whether it converges or diverges, if it is possible, compute its sum:

- i. $\sum_{n=1}^{\infty} \frac{3}{n^2+7n+12}$
- ii. $\sum_{n=4}^{\infty} \frac{10}{n^2-4n+3}$
- iii. $\sum_{n=1}^{\infty} \frac{-2}{n^2+n}$

Solution 1.

$$\begin{aligned} \sum_{n=1}^{\infty} \frac{3}{n^2+7n+12} &= \sum_{n=1}^{\infty} \frac{3}{(n+3)(n+4)} \\ &= \sum_{n=1}^{\infty} \frac{3}{n+3} - \frac{3}{n+4} \quad (\text{By PFD}) \end{aligned}$$

Then its partial sum is

$$\begin{aligned} S_n &= \sum_{k=1}^n \frac{3}{k+3} - \frac{3}{k+4} \\ &= \left[\frac{3}{4} - \frac{3}{5} \right] + \left[\frac{3}{5} - \frac{3}{6} \right] + \dots + \left[\frac{3}{n+2} - \frac{3}{n+3} \right] + \left[\frac{3}{n+3} - \frac{3}{n+4} \right] \\ &= \frac{3}{4} - \frac{3}{n+4} \quad (\text{i.e. Telescopic}) \end{aligned}$$

By definition: $\lim_{n \rightarrow \infty} S_n = \frac{3}{4} - \frac{3}{n+4} = \frac{3}{4}$

Hence: the series converges to $\frac{3}{4}$. □

Solution 2.

$$\begin{aligned} \sum_{n=4}^{\infty} \frac{10}{n^2-4n+3} &= \sum_{n=1}^{\infty} \frac{10}{(n-1)(n-3)} \\ &= \sum_{n=4}^{\infty} \frac{5}{n-3} - \frac{5}{n-1} \quad (\text{By PFD}) \end{aligned}$$

Then its partial sum is

$$\begin{aligned} S_n &= \sum_{k=1}^n \frac{5}{k-3} - \frac{5}{k-1} \\ &= \left[\frac{5}{1} - \frac{5}{3} \right] + \left[\frac{5}{2} - \frac{5}{4} \right] + \left[\frac{5}{3} - \frac{5}{5} \right] + \dots + \left[\frac{5}{n-5} - \frac{5}{n-3} \right] + \left[\frac{5}{n-4} - \frac{5}{n-2} \right] + \left[\frac{5}{n-3} - \frac{5}{n-1} \right] \\ &= 5 + \frac{5}{2} - \frac{5}{n-2} - \frac{5}{n-1} \quad (\text{i.e. Telescopic}) \end{aligned}$$

By definition: $\lim_{n \rightarrow \infty} S_n = 5 + \frac{5}{2} = \frac{15}{2} \Rightarrow$ the series converges to $\frac{15}{2}$. □

Solution 3. This question is left as an exercise to the reader.

Example 9. Prove that the harmonic series

$$\sum_{n=1}^{\infty} \frac{1}{n}$$

diverges.

Approach 1. We proceed with using contradiction.

BWOC: Suppose the harmonic series converges to $H \in \mathbb{R}$

$\Rightarrow$ by definition:

$$\begin{aligned} H &= \sum_{n=1}^{\infty} \frac{1}{n} \\ &= 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \frac{1}{6} + \frac{1}{7} + \frac{1}{8} + \frac{1}{9} + \frac{1}{10} + \dots \\ &\geq 1 + \frac{1}{2} + \frac{1}{4} + \frac{1}{4} + \frac{1}{6} + \frac{1}{6} + \frac{1}{8} + \frac{1}{8} + \frac{1}{10} + \frac{1}{10} + \dots \\ &= 1 + \frac{1}{2} + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots \\ &= \frac{1}{2} + H \end{aligned}$$

Clearly: $H \geq \frac{1}{2} + H \Leftrightarrow 0 \geq \frac{1}{2} \quad \nexists$

Therefore, the harmonic series diverges. □

The proof by contradiction above is seemingly quite “intuitive”. Let’s proceed with a more rigorous approach, the idea for both proofs are the same but this approach illustrates it more theoretical.

However, before we do so, we need to consider a core question that always arise in the study of mathematics: why would we even care about Harmonic series and not the other divergent series? What is so special about this particular divergent series? From our rather shallow understanding of divergent series, it seems like a series diverges only when the it grows out of bound, and this is how Harmonic series intrigues people: it is a series with its terms tending to 0 as n goes to infinity but still diverges

Approach 2. WTS: $\sum_{n=1}^{\infty} \frac{1}{n}$ diverges.

At this point of the course, we do not have that many tools (yet), so let’s proceed as usual by considering how its partial sums behave.

Consider the m -th partial sum: $S_m = \sum_{n=1}^m \frac{1}{n}$, we wish to show that it is unbounded, meaning it can grow arbitrarily large. To analyze its growth, consider the difference between two specific partial sums:

$$S_{2m} - S_m = \sum_{n=m+1}^{2m} \frac{1}{n}$$

Since $\forall n, m+1 \leq n \leq 2m$, we have that

$$\frac{1}{n} \geq \frac{1}{2m}$$

Hence:

$$S_{2m} - S_m = \sum_{n=m+1}^{2m} \frac{1}{n} \geq \sum_{n=m+1}^{2m} \frac{1}{2m} = \frac{1}{2m} \cdot \sum_{n=m+1}^{2m} 1 = m \cdot \frac{1}{2m} = \frac{1}{2}$$

It shows that the sum always increases by at least $\frac{1}{2}$, every time when m doubles. Intuitively:

$$\begin{aligned} S_{2m} &\geq S_m + \frac{1}{2} \\ S_{4m} &\geq S_{2m} + \frac{1}{2} \geq S_m + \frac{1}{2} + \frac{1}{2} \\ S_{8m} &\geq S_{4m} + \frac{1}{2} \geq S_{4m} + \frac{1}{2} \geq S_{2m} + \frac{1}{2} \geq S_m + \frac{1}{2} + \frac{1}{2} + \frac{1}{2} \\ &\dots \end{aligned}$$

Let's rigorously prove it by using induction on m . WTS: $S_m = S_{2^k} = \sum_{n=1}^{2^k} \frac{1}{n} \geq \frac{k}{2} \forall k \geq 1$ (i.e. we set $m = 2^k$ to simplify the proof structure as doubling the terms at each step estimate the growth of the sum more neatly.)

Base case: when $k = 1$:

$$S_{2^1} = S_2 = 1 + \frac{1}{2} = \frac{3}{2} \geq \frac{1}{2}$$

I.H.: Suppose for some $k \geq 1$, $S_{2^k} \geq \frac{k}{2} \Rightarrow$ WTS: $S_{2^{k+1}} \geq \frac{k+1}{2}$

By definition:

$$S_{2^{k+1}} = S_{2^k} + \sum_{n=2^k+1}^{2^{k+1}} \frac{1}{n}.$$

Using the **inductive hypothesis**:

$$S_{2^k} \geq \frac{k}{2}.$$

Now, consider the sum of the next set of terms:

$$\sum_{n=2^k+1}^{2^{k+1}} \frac{1}{n}$$

Since for all n in this range, we have $n \leq 2^{k+1}$, it follows that:

$$\frac{1}{n} \geq \frac{1}{2^{k+1}}$$

There are exactly 2^k terms in this sum, so:

$$\sum_{n=2^k+1}^{2^{k+1}} \frac{1}{n} \geq 2^k \cdot \frac{1}{2^{k+1}} = \frac{1}{2}$$

Thus, we obtain:

$$S_{2^{k+1}} = S_{2^k} + \sum_{n=2^k+1}^{2^{k+1}} \frac{1}{n} \geq \frac{k}{2} + \frac{1}{2} = \frac{k+1}{2}$$

$\therefore m = 2^k \Rightarrow$ as $k \rightarrow \infty$, $2^k = m \rightarrow \infty$, then $S_{2^{k+1}} \rightarrow \infty$ as $k \rightarrow \infty$. Hence, $\lim_{m \rightarrow \infty} S_m = \infty$. □

Approach 3. Another direct approach is via Integral Test. Define $f(n) = \frac{1}{n}$.

$\forall x \in (1, \infty]$, $\frac{1}{x} > 0$ (i.e. it is clearly positive). To see that it is decreasing, we take its derivative: $f'(x) = -\frac{1}{x^2}$, which is negative $\forall x \in [1, \infty)$. Additionally, $f(x)$ is a rational function and is hence continuous on $[1, \infty) \subseteq \text{dom}(f) = \mathbb{R} \setminus \{0\}$. Alternatively, consider differentiability implies continuity.

Then:

$$\begin{aligned} \int_1^\infty \frac{1}{x} dx &= \lim_{A \rightarrow \infty} \int_1^A \frac{1}{x} dx && \text{(By Type-I: infinite integration limit)} \\ &= \lim_{A \rightarrow \infty} (\ln(x) \Big|_1^A) \\ &= \lim_{A \rightarrow \infty} \ln(A) - \ln(1) \\ &= \ln(\infty) - 0 \\ &= \infty && \text{(i.e. it diverges)} \end{aligned}$$

Hence, by the contrapositive of IT, we conclude that the Harmonic series diverges. $\square$

Integral Tests

Example 10. Determine whether the following converges or diverges:

- i. $\sum_{n=0}^\infty \frac{n^3}{n^4+1}$
- ii. $\sum_{n=0}^\infty n \cdot e^{-n}$
- iii. $\sum_{n=2}^\infty \frac{\ln(n)}{n}$

Solution 1. We will apply IT. Define $f(n) = a_n$. As always, we shall first verify the hypothesis.

Increasing: we differentiate our $f(x)$, $f'(x) = \frac{-x^2(x^4-3)}{(x^4+1)^2}$. Note that our denominator is positive, so we will have to analyze the numerator and hoping it is negative. It is clear that the sign depends on the value of x that will make $x^4 - 3 > 0$. We simply obtain $x > \sqrt[4]{3}$. However, this does not cover our entire domain: $[0, \infty)$.

It seems like we ran into a problem, but we can still proceed actually. Bear in mind that series convergence/divergence is a limit process, for most of the time, all we care about is its “tail” behaviour. We would indeed run into a problem if our function is increasing on the whole interval. However, it does not mean that we do not have to do anything - we need to separate the terms from the series that produces an increasing interval and focus on the “well-behaved” interval only.

Note that $\sqrt[4]{3} \approx 1.3$, hence we drop the first two terms in the series and focus on $\sum_{n=2}^\infty \frac{n^3}{n^4+1}$. If this series diverges, then adding a few real-valued terms later will still lead to divergence, same with convergence - I.T. is not supposed to help us to calculate the value it converges to anyways.

Positivity: it is clear that $f(x) > 0 \forall x \in [0, \infty)$.

Continuity: note that $f(x)$ is rational and it's continuous on $[0, \infty) \subseteq \text{dom}(f) = \mathbb{R}$ (i.e. the denom. is never 0).

$$\begin{aligned}
I &= \int_2^{\infty} \frac{x^3}{x^4+1} dx \\
&= \int_{17}^{\infty} \frac{1}{u} \cdot \frac{du}{4} && (\text{let } u = x^4 + 1, \text{ so that } du = 4x^3 dx) \\
&= \lim_{A \rightarrow \infty} \int_{17}^A \frac{1}{4u} dx && (\text{by Type-I}) \\
&= \lim_{A \rightarrow \infty} \frac{1}{4} (\ln |A| - \ln |17|) \\
&= \infty && (\text{i.e. it diverges})
\end{aligned}$$

Since the integral diverges, the series $\sum_{n=0}^{\infty} \frac{n^3}{n^4+1}$ also diverges, by I.T. □

Solution 2. Refer to tutorial notes or left as an exercise for the reader.

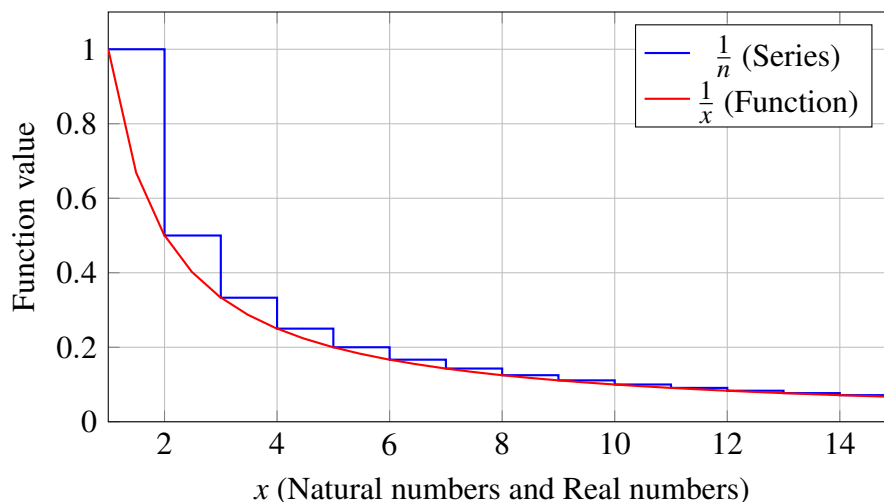
Solution 3. Refer to tutorial notes or left as an exercise for the reader.

Example 11. True or False: If f is a positive, continuous, and decreasing function on $[1, \infty)$ such that $a_n = f(n)$ for all $n \in \mathbb{N}$, then

$$\sum_{n=1}^{\infty} a_n = \int_1^{\infty} f(x) dx.$$

Solution. This is clearly false, consider the harmonic series again:

Comparison of the Harmonic Series and the Continuous Function $1/x$



Note that we have proved the Harmonic series is decreasing, and the plot of Harmonic series is a left Riemann sum. Recall from your W1/W2 material, left Riemann sum with decreasing function overestimates a decreasing function. Hence,

$$\sum_{n=1}^{\infty} a_n > \int_1^{\infty} f(x) dx$$

□

Example 12. Find an example of a convergent geometric series $\sum a_n$ and a divergent geometric series $\sum b_n$ such that the series $\sum \frac{a_n}{b_n}$ converges. Make sure to prove why your examples satisfy the desired criteria.

Solution. Consider $a_n = \frac{1}{2^n}$, $b_n = 2^n$. Clearly, both series are geometric series with a ratio of $\frac{1}{2}$ and 2, respectively, and $a, b = 1$. Then by GST, $\sum a_n$ clearly converges to 2 since the common ratio is less than one, while b_n diverges since the common ratio is larger than or equal to 1.

Now, let's consider their quotient:

$$\sum \frac{a_n}{b_n} = \sum \frac{\frac{1}{2^n}}{2^n} = \sum \frac{1}{2^n \cdot 2^n} = \sum \frac{1}{4^n}$$

Hence, it is still a geometric series but with common ratio $\frac{1}{4} < 1$. Therefore, by GST, the quotient converges. $\square$

Example 13. Find the value(s) of constant c if

$$\sum_{n=2}^{\infty} (1+c)^{-n} = 2.$$

Make sure to fully justify your solution.

Solution. First note that the given series is geometric, by rewriting it

$$\sum_{n=2}^{\infty} (1+c)^{-n} = \sum_{n=0}^{\infty} (1+c)^{-2} \cdot \frac{1}{(1+c)^n}$$

clearly, the common ratio $r = \frac{1}{1+c}$ and $a = \frac{1}{(1+c)^2}$

Because it converges to 2, then by GST:

$$\begin{aligned} \frac{1}{1 - \frac{1}{1+c}} \cdot (1+c)^{-2} &= 2 \Leftrightarrow \frac{1+c}{c} \cdot (1+c)^{-2} = 2 \\ \Leftrightarrow 1 &= 2c(1+c) \\ \Leftrightarrow 1 &= 2c + 2c^2 \\ \Leftrightarrow 0 &= 2c^2 + 2c - 1 \end{aligned}$$

By quadratic formula, we get $c = \frac{-1 \pm \sqrt{3}}{2}$. The remaining task is to use an estimation to show that one of the c is impossible (i.e. both might work, but you will have to check it regardless). Note that $\sqrt{3} > 1$, then $-1 - \sqrt{3} < -2$. Therefore:

$$c = \frac{-1 + \sqrt{3}}{2}$$

is our only solution. $\square$

Example 14. For each of the following series, determine if AST is applicable, or not. If it is applicable, use AST to show convergence/divergence; if not, state the reason why.

i. $\sum_{n=1}^{\infty} \frac{(-1)^{n-10} n^2}{n^3 + n^2 + 4}$

ii. $\sum_{n=1}^{\infty} \frac{(-1)^n n}{n+1}$

iii. $\sum_{n=1}^{\infty} \frac{4n \cos(n\pi)}{2n^2 + 1}$

Example 15. For each of the following series, use CT/LCT to determine whether it converges or diverges:

i. $\sum_{n=1}^{\infty} \frac{(1 - \sin(n))(1 + \sin(n))}{n^2 + 8n + 1}$

ii. $\sum_{n=2}^{\infty} \frac{\sqrt{2 + \cos^2(5n)}}{\sqrt{n^2 - n - 1}}$

iii. $\sum_{n=0}^{\infty} \frac{2^n \sin^2(5n)}{4^n + \cos^2(n)}$